INSTITUTE OF DISTANCE LEARNING

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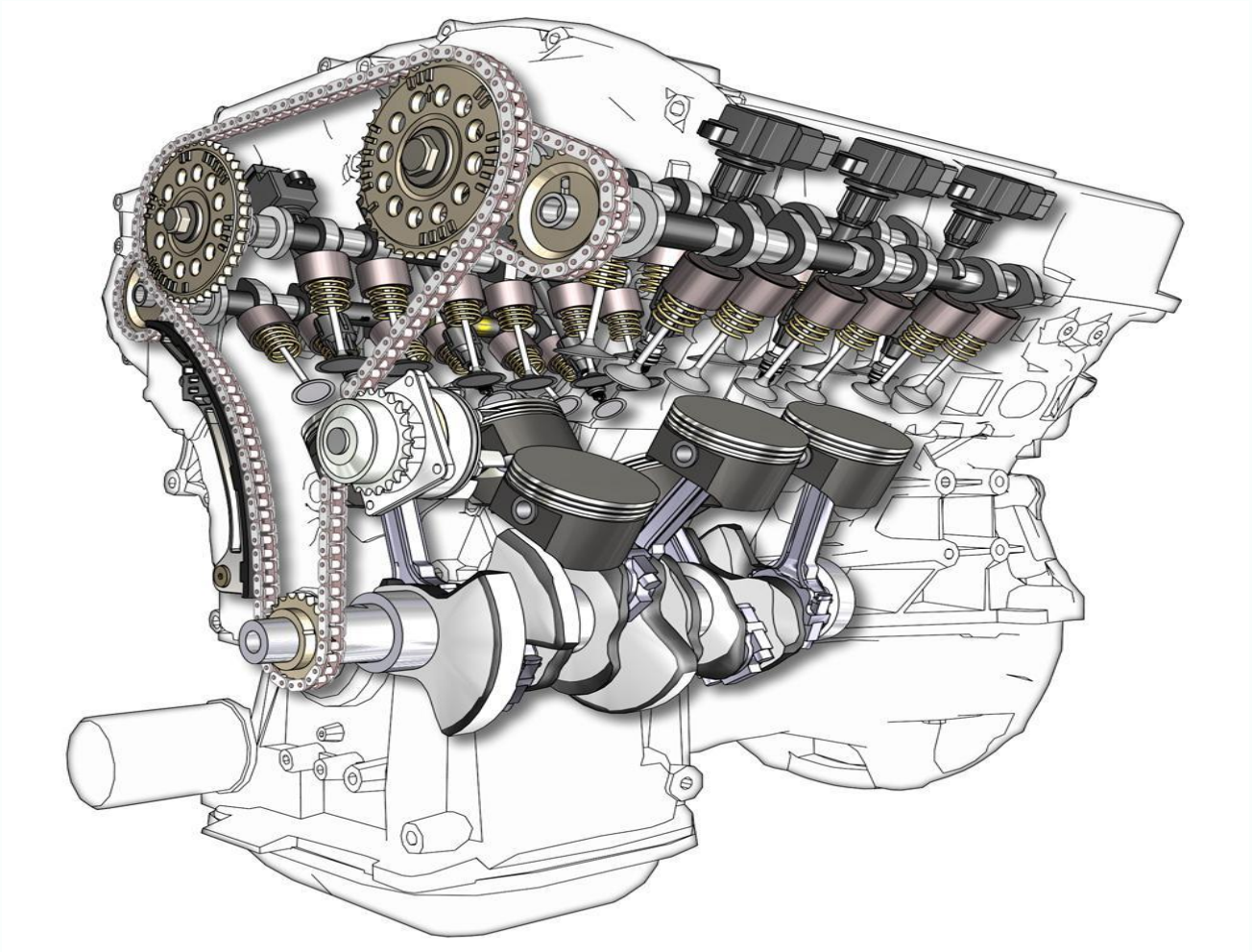
ME 361
DYNAMICS OF MACHINERY

Lecture 6

BALANCING OF RECIPROCATING ENGINES

Mr. Faisal Wahib Adam
Mechanical Engineering Department, KNUST

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BALANCING

- WHAT IS BALANCING ?
- According to ISO definition, balancing is procedure by which the mass distribution of a rotor is checked and if necessary, adjusted in order to ensure that the vibration of the journals and/or force on the bearings at a frequency corresponding to the service speed is within specified limits.
- Everything that rotates needs to be in a state of balance to ensure smooth running when in operation.
- Precision balancing is essential to the manufacture of rotating equipment and to the repair and renovation of installed machines. As machine speeds increase, the effects of unbalance become more detrimental.
- Modern technology allows for accurate balancing to be performed both in the field and in the workshop.

- **BENEFITS OF BALANCED SYSTEMS**
- There are many documented "good consequences" and benefits associated with a well balanced, smooth running rotor. Included among these are:
- **Minimize vibration**
- **Minimize noise**
- **Minimize structural stress**
- **Minimize operator fatigue and annoyance**
- **Increase machine life**
- **Increase bearing life**
- **Increase product quality**
- **Increase personnel safety**
- **Increase productivity**
- **Lower operating costs**

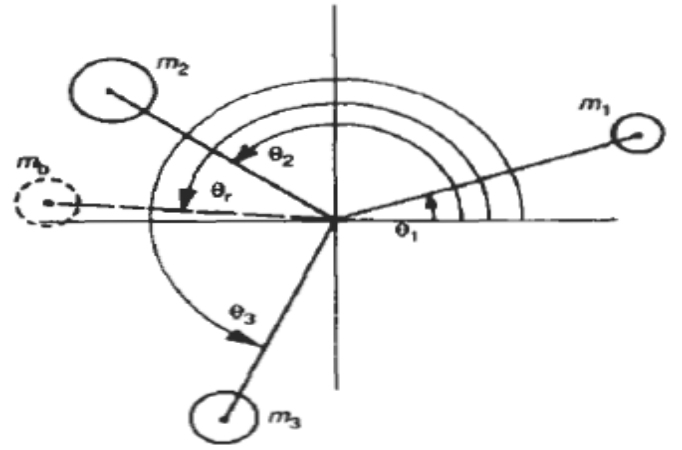
BALANCING

- ✓Balancing of rotating engines
  - Same plane (Static)
  - Different planes (Dynamic)
- ✓Balancing of reciprocating engines
  - Inline
  - Radial
    - ➢ V-Engines
    - ➢ W-Engines etc.

Review-Balancing of rotating masses

Same plane (Static)

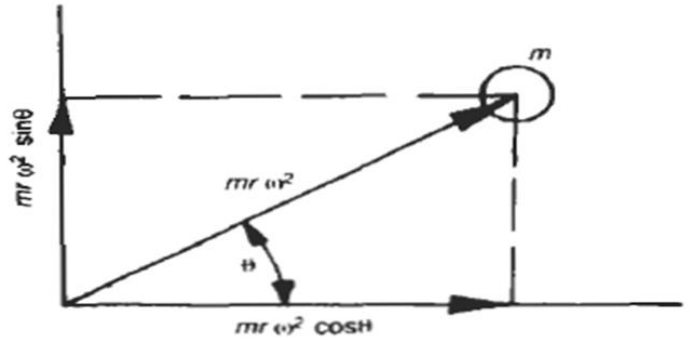
Balancing of rotating components is of extreme importance, especially in the *case* of high-speed machinery. Lack of balance may be due to masses in one plane or masses in different planes some distance apart. The method of balancing is given.



The forces are: $m_1\omega^2 r_1$, $m_2\omega^2 r_2$, etc. These are resolved into vertical and horizontal components:

$$F_V = m_1\omega^2 r_1 sin\theta_1 + m_2\omega^2 r_2 sin\theta_2 + \cdots$$
$$F_H = m_1\omega^2 r_1 cos\theta_1 + m_2\omega^2 r_2 cos\theta_2 + \cdots$$

Resultant force, $F = \sqrt{{F_V}^2 + {F_H}^2}$ at an angle to horizontal axis $\theta = \tan^{-1}\left(\frac{F_V}{F_H}\right)$



Review-Balancing of rotating masses Same plane (Static)

EXAMPLE

Four masses m_1, m_2, m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45°, 75° and 135°. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m.

$$\sum F_V = 200 \times 0.2sin0 + 300 \times 0.15sin45 + 240 \times 0.25sin120 + 260 \times 0.3sin255$$

$$\sum F_V = \ 8.5 \text{ kg m}$$

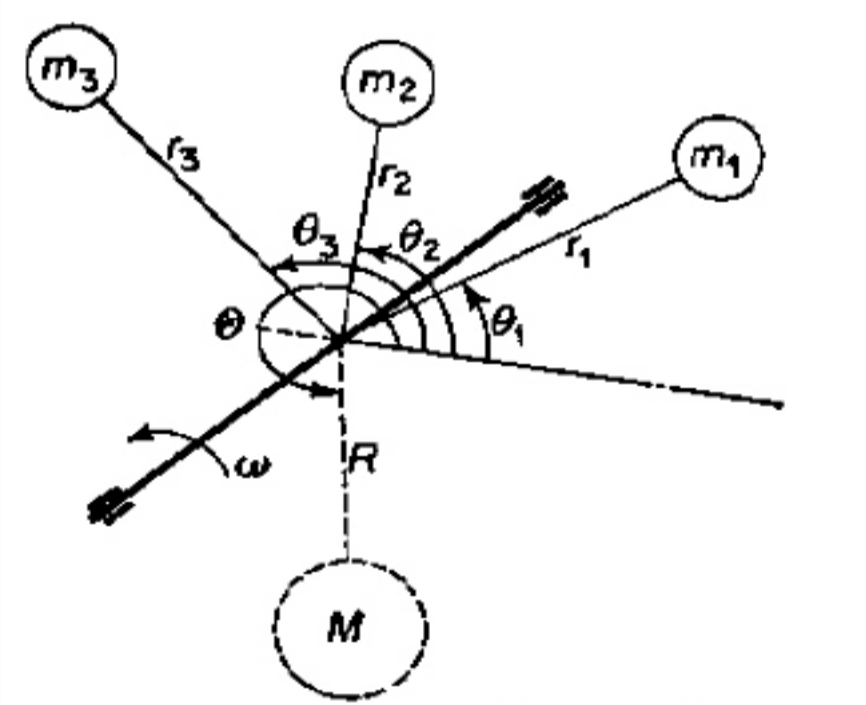
$$\sum F_H = 200 \times 0.2cos0 + 300 \times 0.15cos45 + 240 \times 0.25cos120 + 260 \times 0.3cos255$$

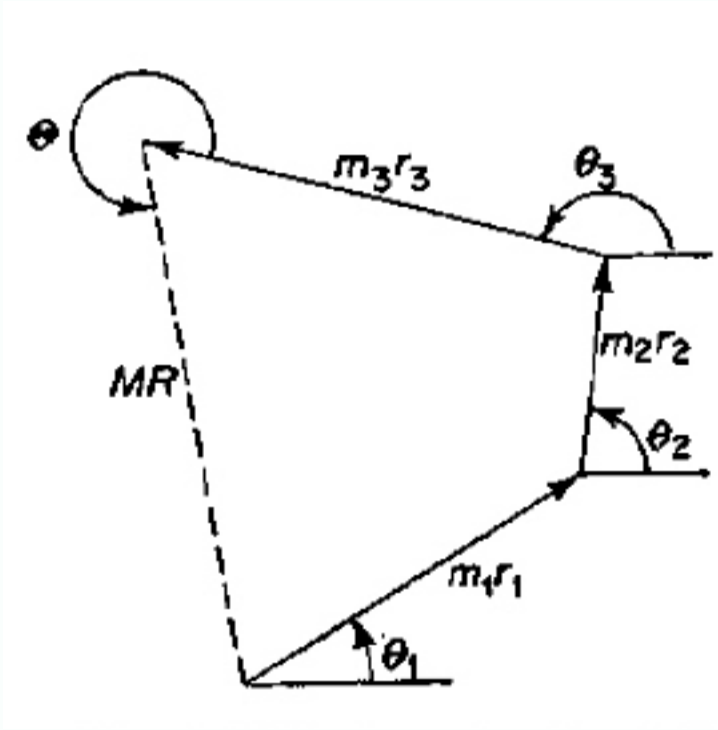
$$\sum F_H = \ 21.6 \text{ kg m}$$

Unbalance force, $F = \sqrt{\sum F_V{}^2 + \sum F_H{}^2} = 23.2$ kg m

$$\theta = \tan^{-1}\left(\frac{8.5}{21.6}\right) = 21.48°$$

Graphical Method





Review-Balancing of rotating masses Different planes (Dynamic)

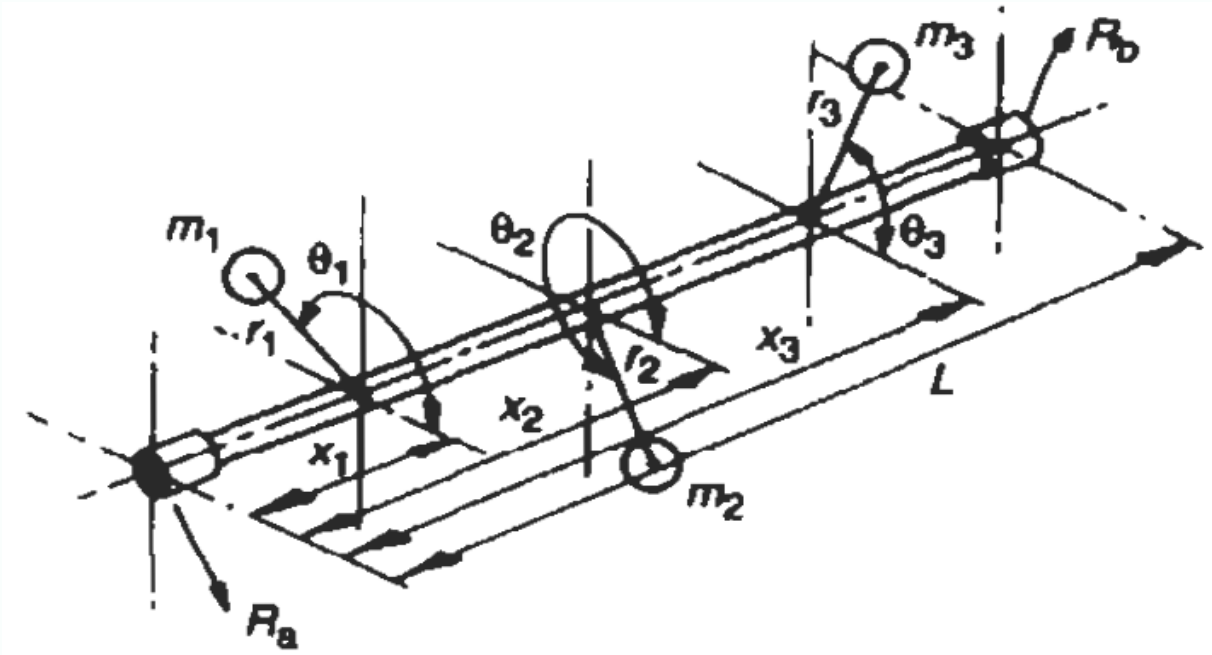
Method of *balancing* Complete dynamic balance *is* achieved by introducing forces equal and opposite to R_a, and R_b. In practice, balancing is carried out at planes a short distance from the bearings.

$$\sum F_V = \sum_{i=1}^{i=n} m_i \omega^2 r_i sin\theta_i = 0$$

$$\sum F_H = \sum_{i=1}^{i=n} m_i \omega^2 r_i cos\theta_i = 0$$

$$\sum M_a = \sum_{i=1}^{i=n} m_i \omega^2 r_i x_i sin\theta_i = 0$$

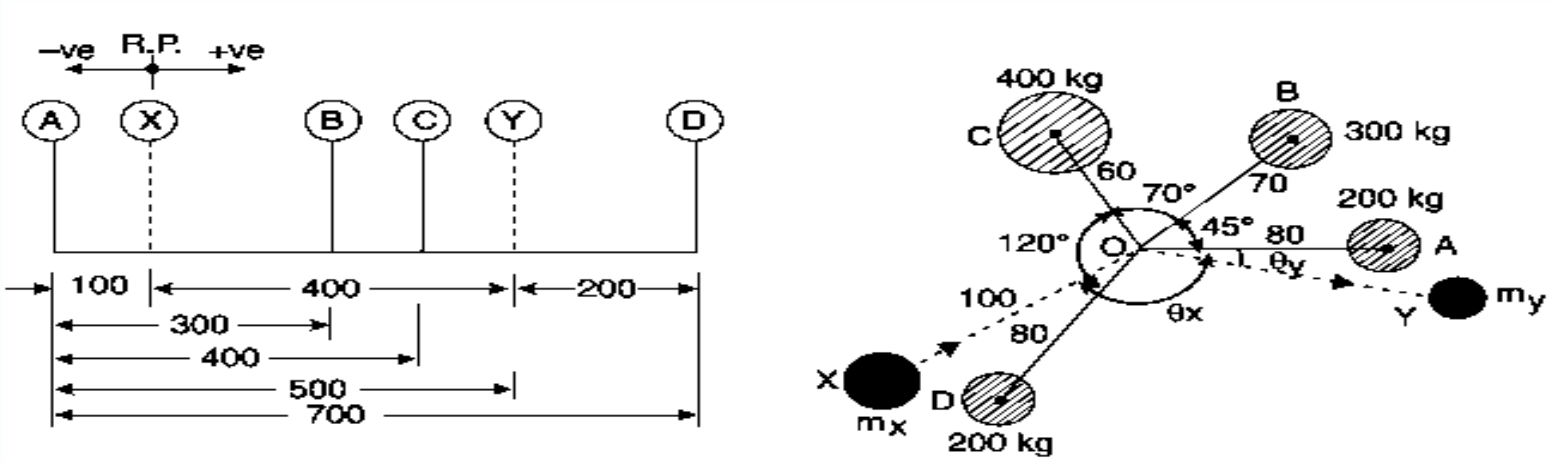
$$\sum M_b = \sum_{i=1}^{i=n} m_i \omega^2 r_i x_i cos\theta_i = 0$$



Review-Balancing of rotating masses Different planes (Dynamic)

EXAMPLE

A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.



All dimensions in mm.

(*a*) Position of planes. (*b*) Angular position of masses.

Review-Balancing of rotating masses Different planes (Dynamic)

Plane	*M(kg)*	*r(m)*	*(M.r) kg-m*	*l (m)*	*(M.r.l) kg-m²*	*θ°*	*(M.r) cosθ*	*(M.r) sinθ*	*(M.r.l) cosθ*	*(M.r.l) sinθ*
A	200	0.08	16	- 0.1	- 1.6	0	16	0	-1.6	0
X(R.P.)	mx	0.1	0.1 mx	0	0	θ_X	0.1 mx cosθ_X	0.1mxcosθ_X	0	0
B	300	0.07	21	0.2	4.2	45	14.85	14.85	2.97	2.97
C	400	0.06	24	0.3	7.2	115	-10.14	21.75	-3.04	6.53
Y	my	0.1	0.1 my	0.4	0.04 my	θ_Y	0.1mycosθ_Y	0.1mysinθ_Y	0.04mycosθ_Y	0.04mysinθ_Y
D	200	0.08	16	0.6	9.6	235	-9.18	-13.11	-5.51	-7.86

$$\sum mrlcos\theta = 0.04\text{mycos}\theta_Y - 7.18 = 0;$$

$$\sum mrlsin\theta = 0.04\text{mysin}\theta_Y + 1.67 = 0;$$

$$\text{mycos}\theta_Y = 179.5;$$

$$\text{mysin}\theta_Y = -41.75;$$

$$my = \sqrt{179.5^2 + 41.75^2} = 184.29\ kg < -13.09°;$$

$$\sum mrcos\theta = 0.1\text{mxcos}\theta_X + 29.48 = 0;$$

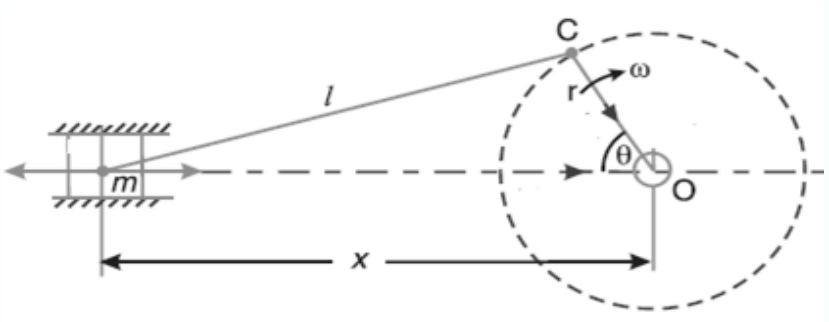
$$\sum mrsin\theta = 0.1\text{mxsin}\theta_X - 19.32 = 0;$$

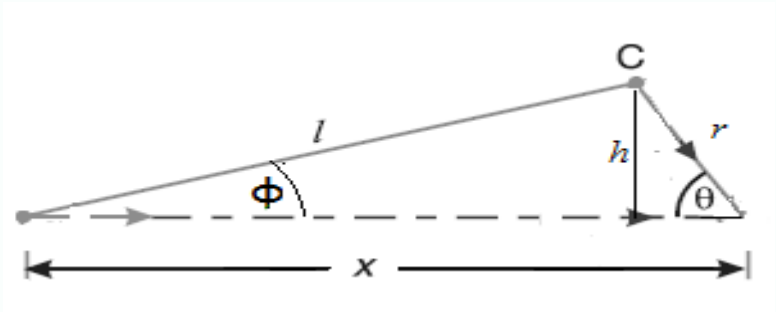
$$\text{mxcos}\theta_x = -294.8;$$

$$\text{mxsin}\theta_X = 193.2;$$

$$mx = \sqrt{294.8^2 + 193.2^2} = 352.5\ kg < 146.76°;$$

Balancing of inline (reciprocating) engines





$$h = r\sin\theta = l\sin\phi$$

$$x = r\cos\theta + l\cos\phi$$

$$\cos^2\phi + \sin^2\phi = 1 \Rightarrow \cos\phi = \sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

$$\therefore\ x = r\cos\theta + l\sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

$$recall\ ;(1+x)^n = 1 + xn + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + x^n$$

$$x = r\cos\theta + l\left\{1 - \frac{1}{2}\left(\frac{r}{l}\sin\theta\right)^2 + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{r}{l}\sin\theta\right)^4 + \ldots - \frac{r}{l}\sin\theta\right\}$$

$$\dot{x} = -r\omega\sin\theta - \frac{r^2}{2l}\omega\sin 2\theta\ ;\ taking\ \text{the first two items of the series}$$

$$\ddot{x} = -r\omega^2\left(\cos\theta + \frac{1}{n}\cos 2\theta\right)$$

$\therefore$ the unbalance (shaking) force on one cylinder is

$$F = m\ddot{x} = m\omega^2 r\left(\cos\theta + \frac{1}{n}\cos 2\theta\right);\ \text{where}\quad n = \frac{l}{r}$$

Balancing of inline (reciprocating) engines cont'd…

For more than one cylinder

$$F = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \cos(\theta + \phi_i) + \frac{1}{n} \cos(2\theta + 2\phi_i) \right\};$$

$\phi_1 = 0$

The primary force is

$$F_p = \omega^2 \sum_{i=1}^{i=z} m_i r_i \cos(\theta + \phi_i)$$

And the secondary force is

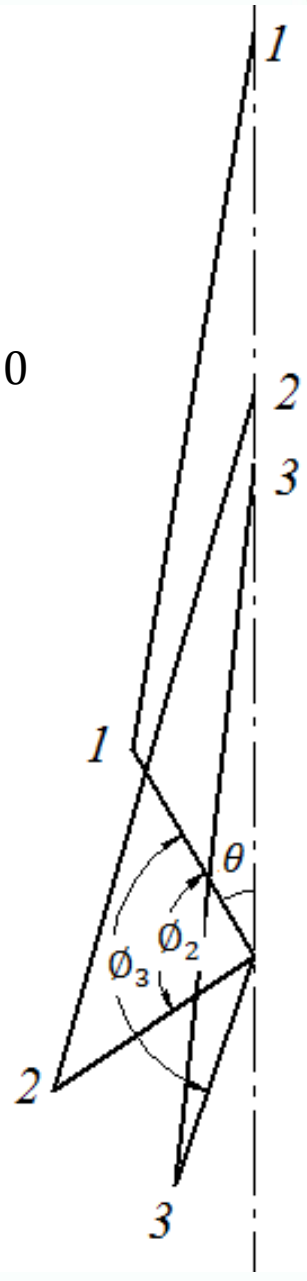
$$F_S = \frac{\omega^2}{n} \sum_{i=1}^{i=z} m_i r_i \cos(2\theta + 2\phi_i)$$

The primary force is

$$C_p = \omega^2 \sum_{i=1}^{i=z} m_i r_i x_i \cos(\theta + \phi_i)$$

And the secondary couple is

$$C_S = \frac{\omega^2}{n} \sum_{i=1}^{i=z} m_i r_i x_i \cos(2\theta + 2\phi_i)$$



Balancing of inline (reciprocating) engines cont'd…

Expanding the above equations gives the following conditions for primary and secondary balancing

$$\sum_{i=1}^{i=z} m_i r_i \sin\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i \cos\phi_i = 0, \text{conditions for primary force balance}$$

$$\sum_{i=1}^{i=z} m_i r_i \sin 2\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i \cos 2\phi_i = 0, \text{conditions for secondary force balance}$$

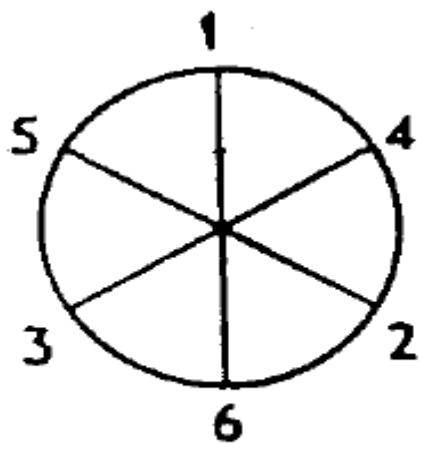
$$\sum_{i=1}^{i=z} m_i r_i x_i \sin\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i x_i \cos\phi_i = 0, \text{conditions for primary couple balance}$$

$$\sum_{i=1}^{i=z} m_i r_i x_i \sin 2\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i x_i \cos 2\phi_i = 0, \text{conditions for secondary couple balance}$$

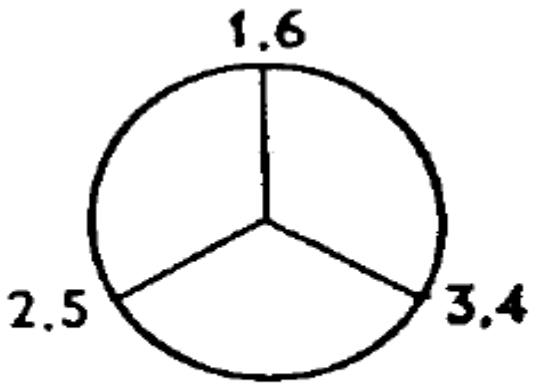
EXAMPLE

The six cylinders of a single-acting, two-stroke cycle Diesel engine are pitched 1 m apart and the cranks are spaced at 60° intervals. The crank length is 300 mm and the ratio of connecting rod to crank is 4.5. The reciprocating mass per line is 1.35 Mg and the rotating mass is 1 Mg. The speed is 200 rev/min. Show with regard to primary and secondary; balance, that the firing order 1-5-3-6-2-4 gives unbalance in primary moment only, and the order 1-4-5-2-3-6 gives secondary moment unbalance only. Compare the maximum values of these moments, evaluating them with respect to the central plane of the engine.

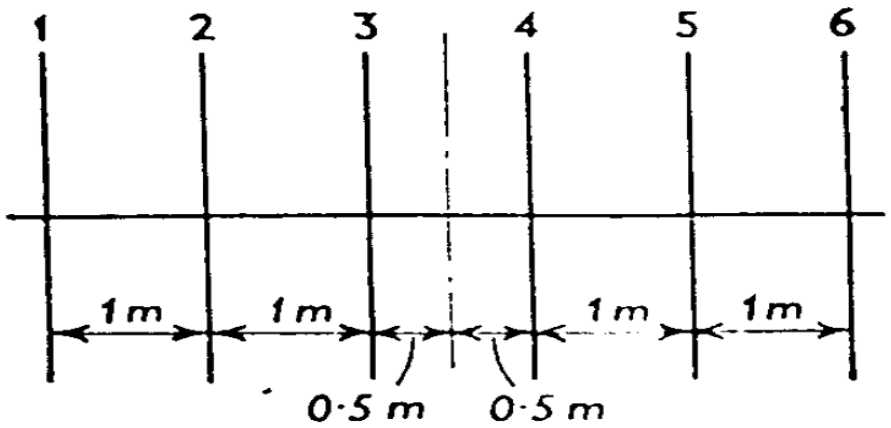
Firing Order 1-5-3-6-2-4.



Primary cranks



Secondary cranks



Graphical Method cont'd…

Plane	R (kg)	M (kg)	r (m)	Rr (kg m)	$(R + M)r$ (kg m)	x (m)	Rrx (kg m²)	$(R + M)rx$ (kg m²)
1	1350	1000	0·3	405	705	−2·5	−1012·5	−1762·5
2	1350	1000	0·3	405	705	−1·5	−607·5	−1057·5
3	1350	1000	0·3	405	705	−0·5	−202·5	−352·5
4	1350	1000	0·3	405	705	+0·5	+202·5	+352·5
5	1350	1000	0·3	405	705	+1·5	+607·5	+1057·5
6	1350	1000	0·3	405	705	+2·5	+1012·5	+1762·5

From the crank symmetry, the primary and secondary forces are balanced. From the primary couple polygon, Fig. 7.11, using figures from the $(R + M)rx$ column, the closing side, 06, represents 2430 kg m².

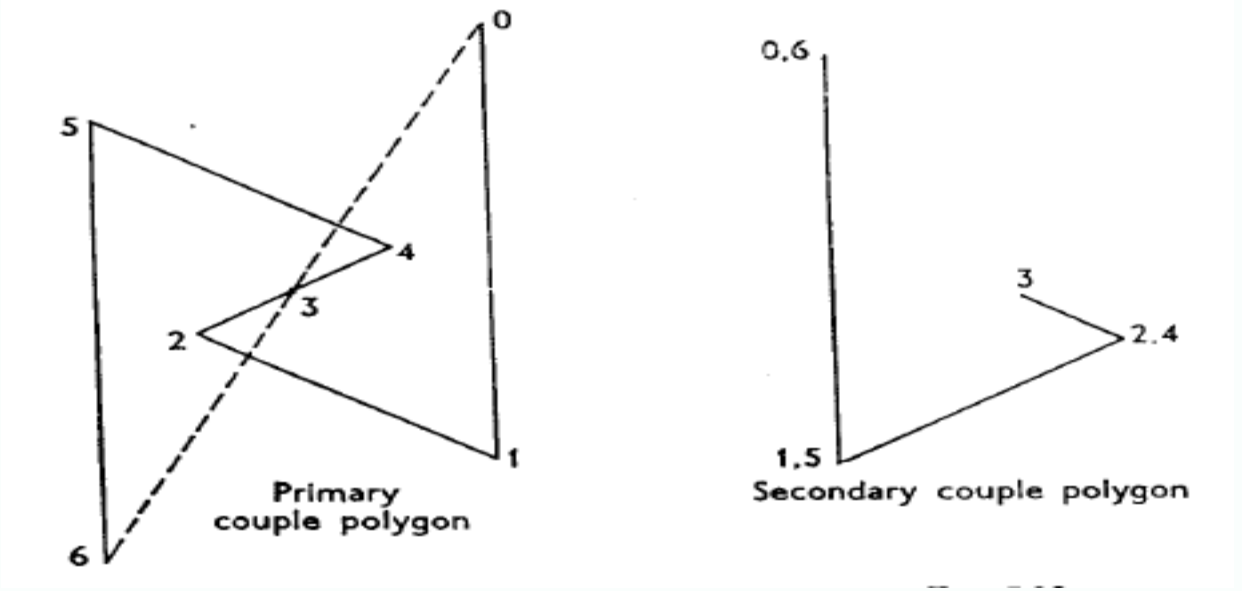
∴ maximum unbalanced primary couple

$$= 2430 \times \left(200 \times \frac{2\pi}{60}\right)^2$$

$$= 1\,065\,000 \text{ N m} = \underline{1{\cdot}065 \text{ MN m}}^*$$

The secondary couple polygon, Fig. 7.12, is a closed figure, so that there is no unbalanced secondary couple.

Graphical Method cont'd…



EXAMPLE

$$\sum sin\emptyset_i = sin0 + sin60 + sin120 + sin180 + sin240 + sin300 = 0$$

$$\sum cos\emptyset_i = cos0 + cos60 + cos120 + cos180 + cos240 + cos300 = 0$$

$$\sum sin2\emptyset_i = sin0 + sin120 + sin240 + sin360 + sin480 + sin600 = 0$$

$$\sum cos2\emptyset_i = cos0 + cos120 + cos240 + cos360 + cos480 + cos600 = 0$$

Therefore the primary and secondary forces are balanced

$$\sum x_i sin\emptyset_i = -2.5sin0 + 1.5sin60 - 0.5sin120 + 2.5sin180 - 1.5sin240 + 0.5sin300 = \sqrt{3}$$

$$\sum x_i cos\emptyset_i = -2.5cos0 + 1.5cos60 - 0.5cos120 + 2.5cos180 - 1.5cos240 + 0.5cos300 = -3$$

Primary couple is not balanced

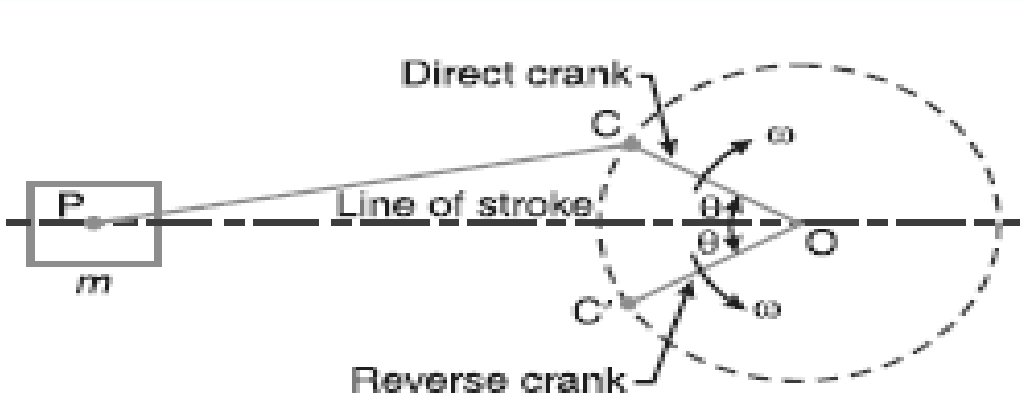
$$\sum x_i sin2\emptyset_i = -2.5sin0 + 1.5sin120 - 0.5sin240 + 2.5sin360 - 1.5sin480 + 0.5sin600 = 0$$

$$\sum x_i cos2\emptyset_i = -2.5cos0 + 1.5cos120 - 0.5cos240 + 2.5cos360 - 1.5cos480 + 0.5cos600 = 0$$

Secondary couples are balanced

Unbalance couple, $C = -(m + M)\omega^2 r\left(\sqrt{3}sin\theta + 3cos\theta\right) = 1.07\cos(\theta - 30^o)\, MNm$

Direct and Reverse Cranks



- Consider the diagram above, if the masses at C should be made equivalent to m/2
- Resolving the horizontal components of this force system gives

$$F_H = \frac{m}{2}\omega^2 r\cos\theta \times 2 = m\omega^2 r\cos\theta$$

Direct and Reverse Cranks cont'd…

- **The vertical component**

$$F_V = \frac{m}{2}\omega^2 r \sin\theta - \frac{m}{2}\omega^2 r \sin\theta = 0$$

- **The secondary vertical force**

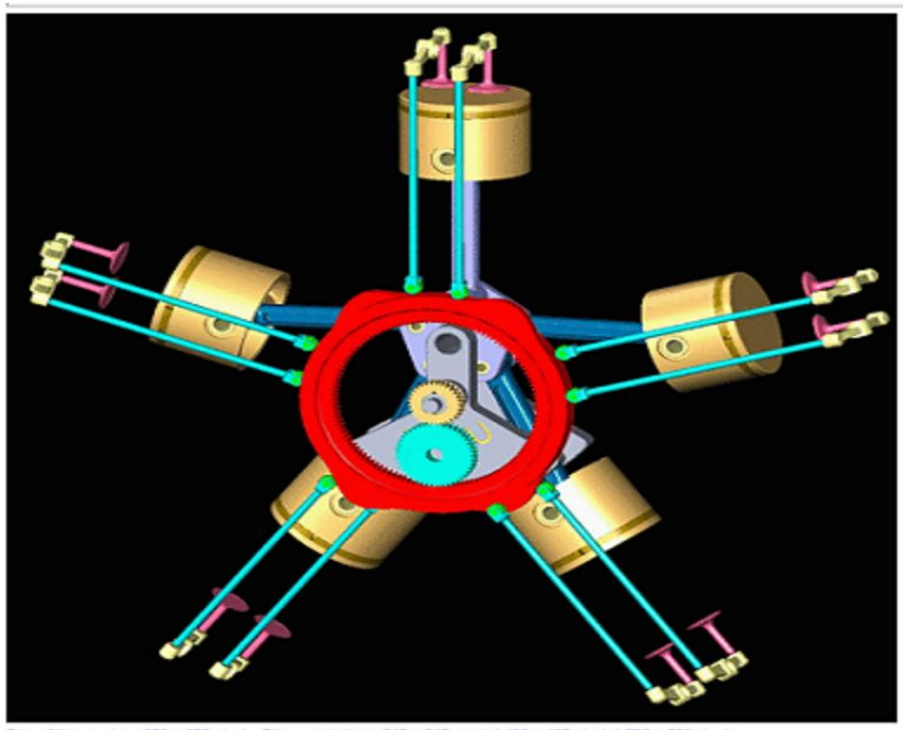
$$F_H = \frac{1}{n} m\omega^2 r \cos 2\theta$$

- **Similarly, the vertical component of the secondary force is zero**

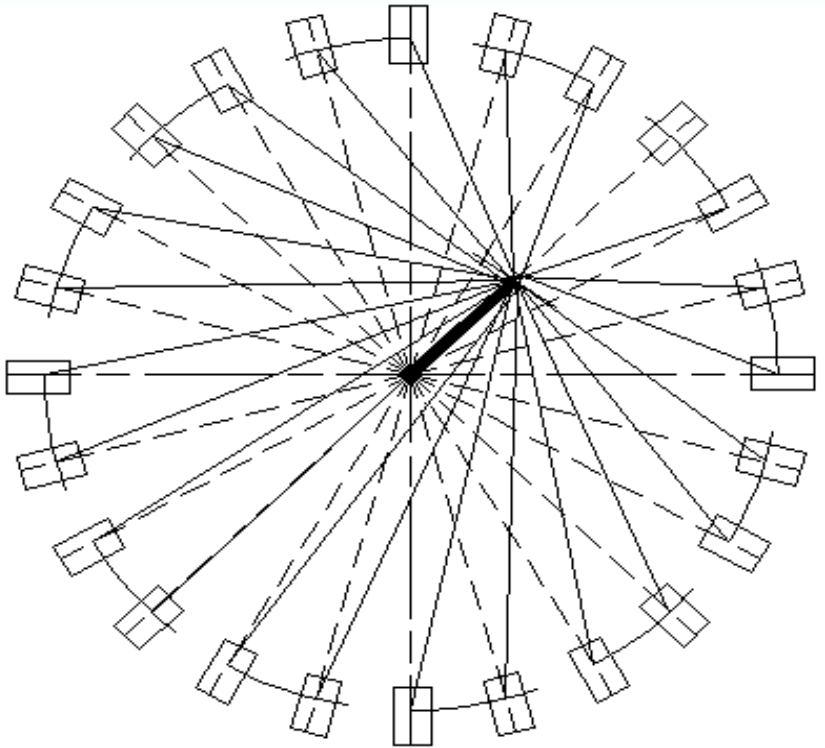
RADIAL ENGINES

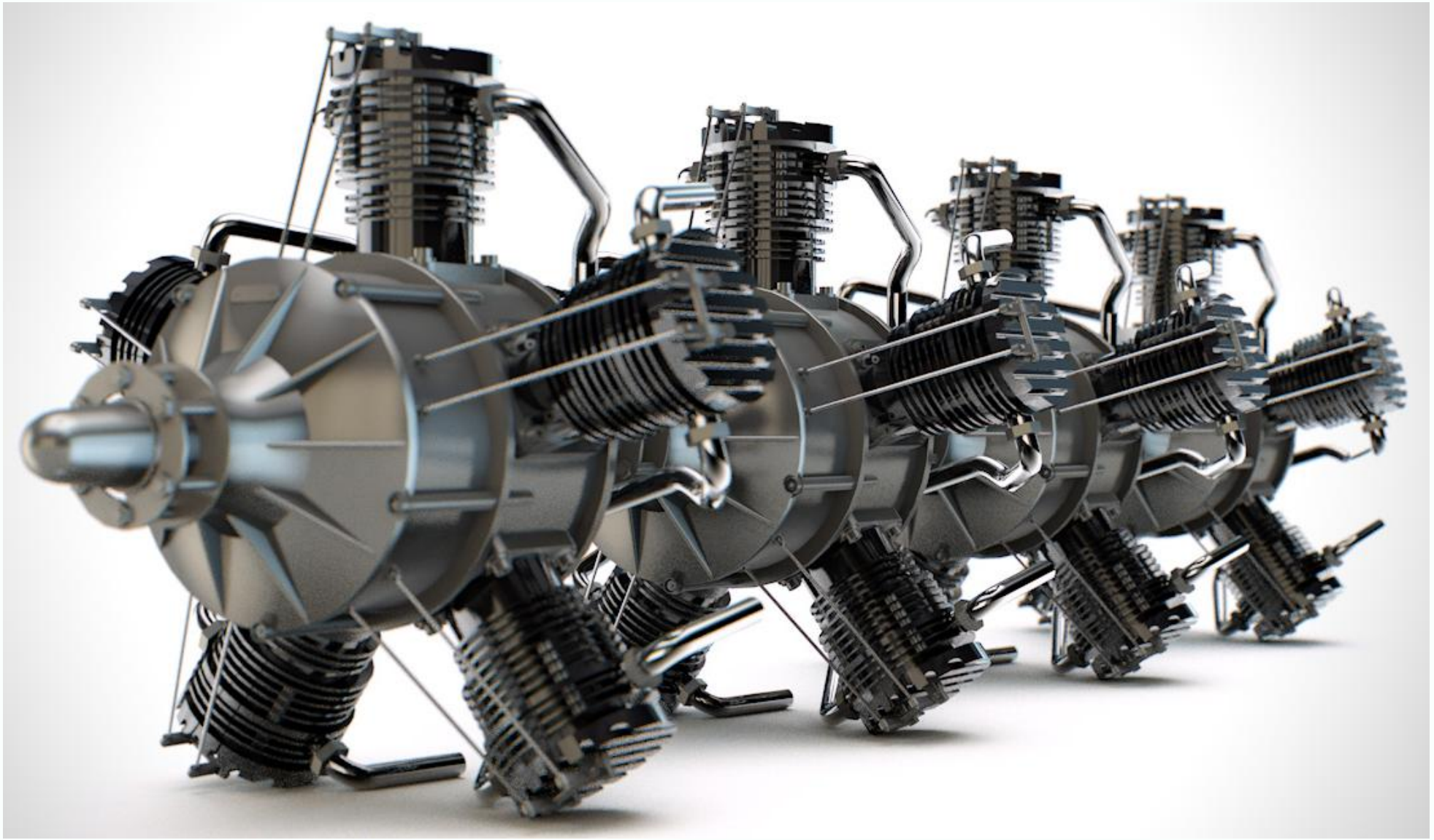
The radial engine is a reciprocating type internal combustion engine configuration in which the cylinders point outward from a central crankshaft. The radial configuration was very commonly used in aircraft engines before turbine engines became predominant.

R-5



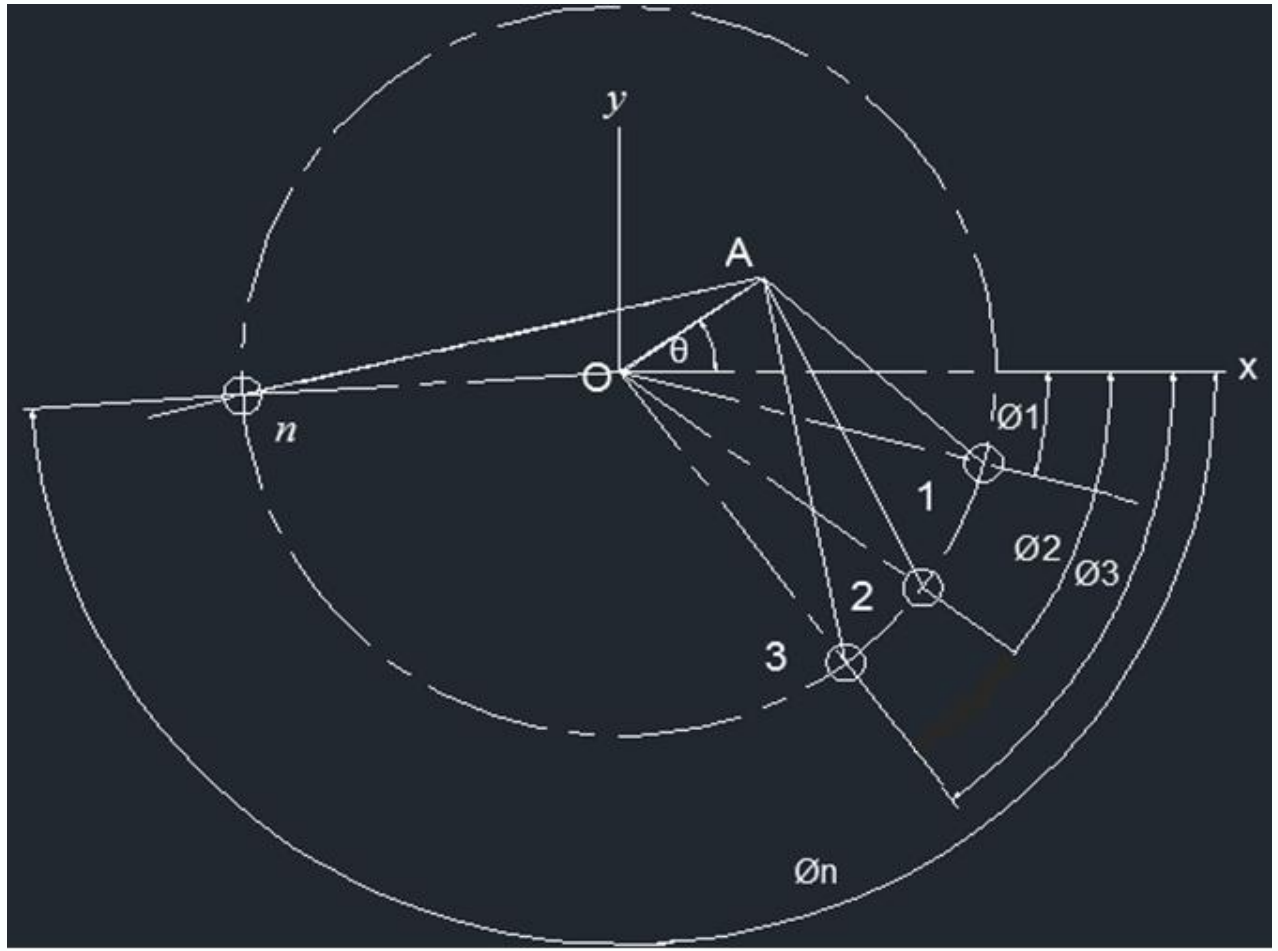
R-24







Balancing of radial engines



Balancing of radial engines cont'd…

Total Horizontal Primary Force

$$F_{px} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \{\cos(\theta + \phi_i)\cos\phi_i\};$$

Total Vertical Primary Force

$$F_{py} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \{\cos(\theta + \phi_i)\sin\phi_i\};$$

Resultant Primary Force

$$R_p = \sqrt{F_x^2 + F_y^2}$$

Balancing of radial engines cont'd…

Total Horizontal Secondary Force

$$F_{sx} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \frac{1}{n} \cos(2\theta + 2\phi_i) \cos\phi_i \right\};$$

Total Vertical Secondary Force

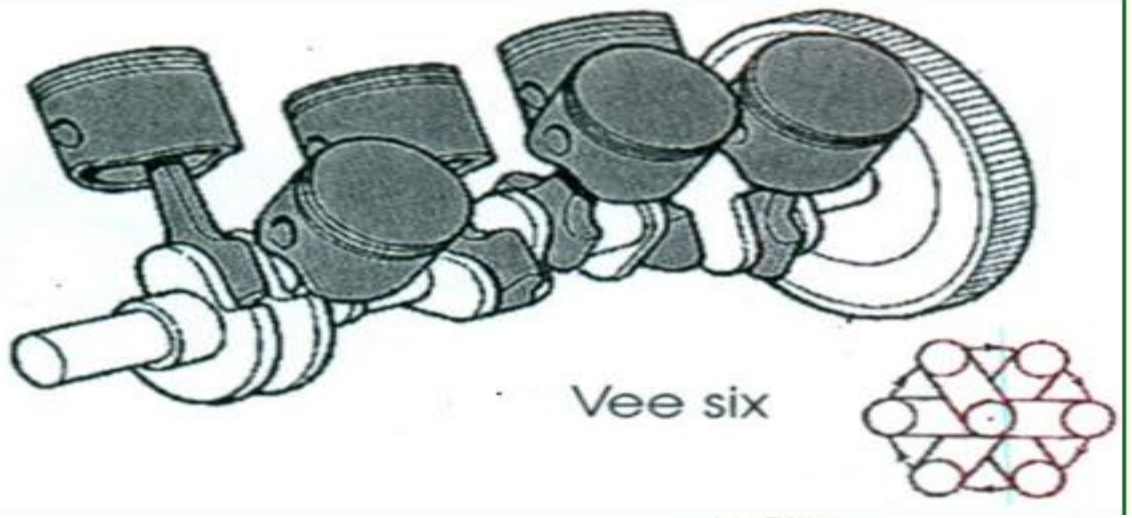
$$F_{sy} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \frac{1}{n} \cos(2\theta + 2\phi_i) \sin\phi_i \right\};$$

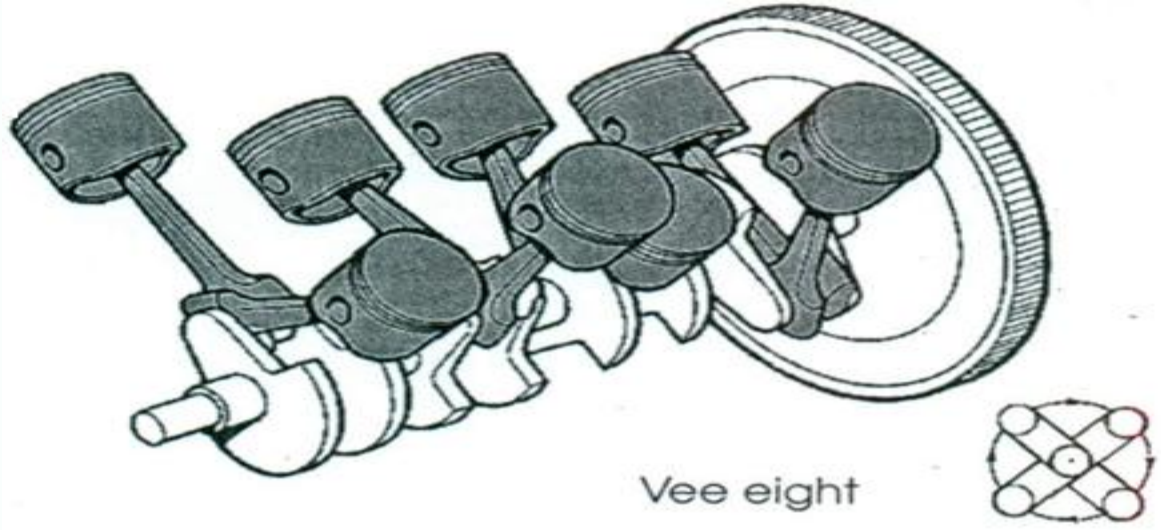
Resultant Secondary Force

$$R_s = \sqrt{F_x^2 + F_y^2}$$

V-ENGINES

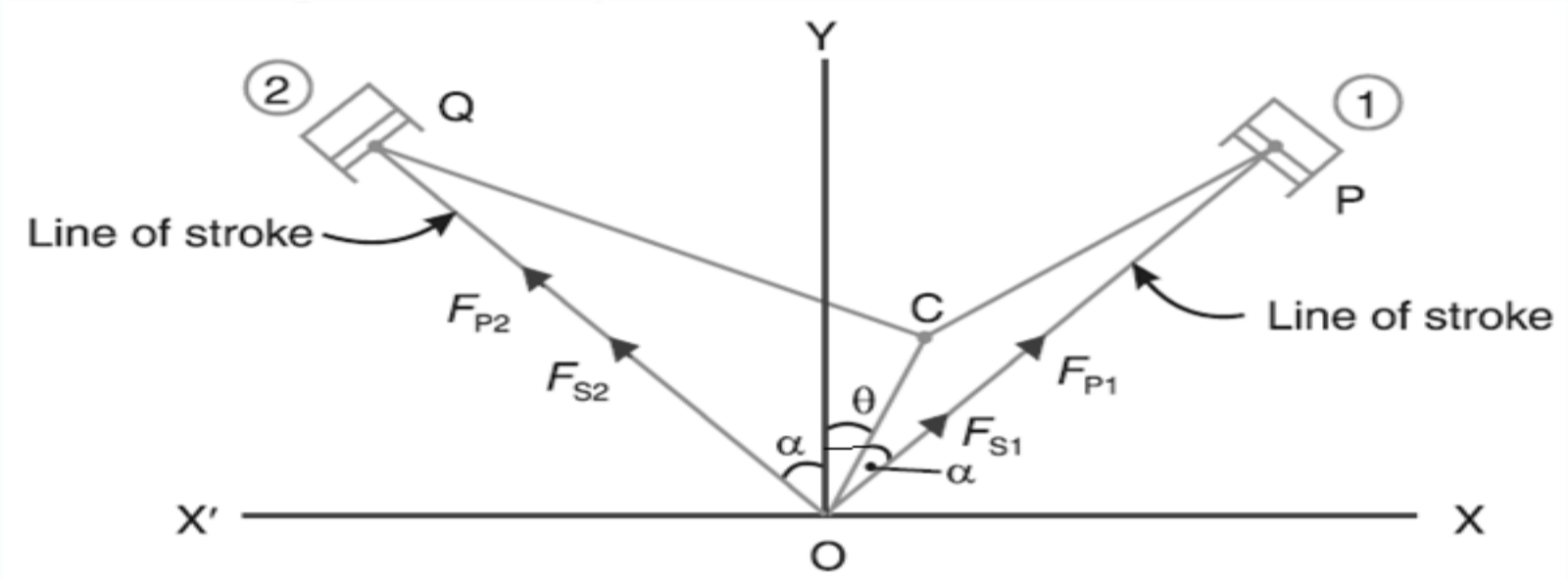
The V-engine configuration is made up of two inline configurations with one on each bank, as shown. A V-6 engine is a six-cylinder engine which essentially consists of two three-cylinder inline engines on a common crankshaft. Similarly, a V-8 engine consists of two-cylinder inline engines on a common crankshaft. In V configurations, the bank angle introduces additional phase shift of the forces.





Balancing of V-Engines

Consider a two-cylinder engine with one cylinder in each bank and with both sharing a common crank throw, and each cylinder inclined at an angle α to the vertical. By letting the y-axis be the central axis of the banks, the x-axis being the horizontal, with angle θ measured from the piston axis, the shaking force along x and y can be calculated.



Balancing of V-Engines

$$Total \quad \text{primary ve} \quad \text{rtical} \quad \text{force}$$

$$F_V = m\omega^2 r\cos(\alpha - \theta)\cos\alpha + m\omega^2 r\cos(\alpha + \theta)\cos\alpha$$

$$F_V = 2m\omega^2 r\cos^2\alpha\cos\theta$$

$$Total \quad \text{primary} \quad \text{horizontal} \quad \text{force}$$

$$F_H = m\omega^2 r\cos(\alpha - \theta)\sin\alpha - m\omega^2 r\cos(\alpha + \theta)\sin\alpha$$

$$F_H = 2m\omega^2 r\sin^2\alpha\sin\theta$$

$$\text{Resultant} \quad \text{primary} \quad \text{force}$$

$$F_{PR} = \sqrt{F_H^2 + F_V^2}$$

$$F_{PR} = 2m\omega^2 r\sqrt{(\sin^2\alpha\sin\theta)^2 + (\cos^2\alpha\cos\theta)^2}$$

Balancing of V-Engines

Similarly the secondary unbalance forces are

$$F_H = \frac{2 m \omega^2 r}{n} (\sin \alpha \sin 2\alpha \sin 2\theta)$$

$$F_V = \frac{2 m \omega^2 r}{n} (\cos \alpha \cos 2\alpha \cos 2\theta)$$

$$F_{SR} = \frac{2 m \omega^2 r}{n} \sqrt{(\sin \alpha \sin 2\alpha \sin 2\theta)^2 + (\cos \alpha \cos 2\alpha \cos 2\theta)^2}$$

Balancing of V-Engines cont'd…

For V - Engines with more than two cylinders

Total primary force on the left bank

$$F_{PL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos \beta_i \cos(\theta + \alpha) + \sin \beta_i \sin(\theta + \alpha)\}$$

Total primary force on the right bank

$$F_{PR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos \beta_i \cos(\alpha - \theta) + \sin \beta_i \sin(\alpha - \theta)\}$$

Total secondary force on the left bank

$$F_{SL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\theta + 2\alpha) + \sin 2\beta_i \sin(2\theta + 2\alpha)\}$$

Total secondary force on the right bank

$$F_{SR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\alpha - 2\theta) + \sin 2\beta_i \sin(2\alpha - 2\theta)\}$$

where β_i is the 'i' th angle a crank makes with a reference.

Example 1

The reciprocating mass per cylinder in a 60° V-twin engine is 1.5 kg. The stroke and connecting rod length are 100 mm and 250 mm respectively. If the engine runs at 2500 rpm., determine the maximum and minimum values of the primary and secondary forces. Also find out the crank position corresponding to these values.

Solution (example 1)-analytical method

Given 2α = 60° or α = 30°, m = 1.5 kg ; Stroke = 100 mm or r = = 50 mm ; l = 250 mm ; N = 250 r.p.m. or ω = 2 π × 2500 / 60 = 261.8 rad/s

Maximum and minimum values of primary forces

We know that the resultant primary force,

$$F_{PR} = 2 m \omega^2 r \sqrt{(\sin^2 \alpha \sin \theta)^2 + (\cos^2 \alpha \cos \theta)^2}$$

$$F_{PR} = 2 m \omega^2 r \sqrt{(\frac{3}{4} \cos \theta)^2 + (\frac{1}{4} \sin \theta)^2}$$

The primary force is maximum, when θ = 0°. Therefore, **maximum primary force is**

$$F_{PR} = 2 \times 1.5 \times (261.8)^2 \times 0.05 \sqrt{(\frac{3}{4})^2} = 7.711 \ kN$$

Solution (example 1) cont'd…

The primary force is minimum, when θ = 90°. Therefore ***the minimum primary force is***

$$F_{PR} = 2 \times 1.5 \times (261.8)^2 \times 0.05 \sqrt{(\frac{1}{4})^2} = 2.570 \quad kN$$

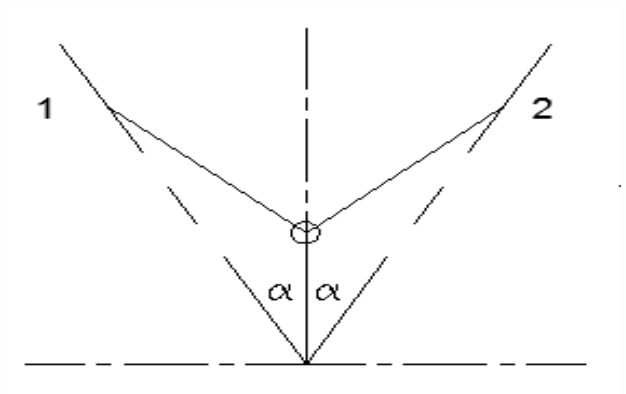
Maximum and minimum values of secondary forces

$$F_{SR} = \frac{2 m \omega^2 r}{n} \sqrt{(\sin \alpha \sin 2\alpha \sin 2\theta)^2 + (\cos \alpha \cos 2\alpha \cos 2\theta)^2}$$

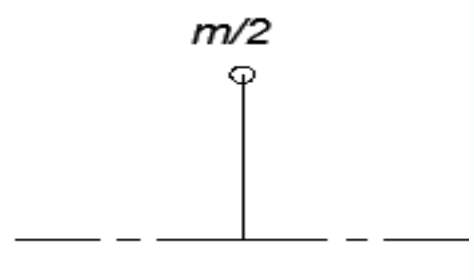
$$F_{SR} = \frac{2 m \omega^2 r}{n} \sqrt{(\frac{1}{2} \times \frac{\sqrt{3}}{2} \sin 2\theta)^2 + (\frac{1}{2} \times \frac{\sqrt{3}}{2} \cos 2\theta)^2}$$

$$F_{SR} = \frac{\sqrt{3} \times 1.5 \times 261.8^2 \times 0.05}{2 \times (0.25 / 0.05)} = 890.3 \text{ N}$$

Graphical method (example 1, Solutic

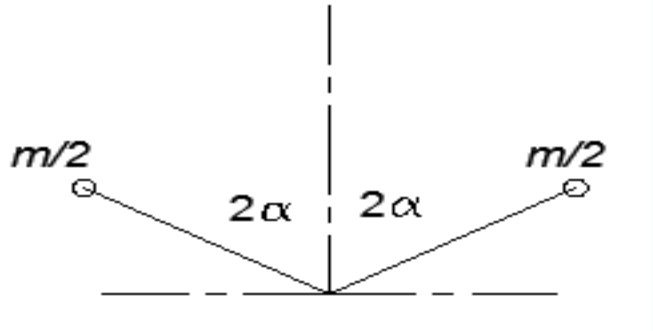


Arrangement of the Cylinders and Crank



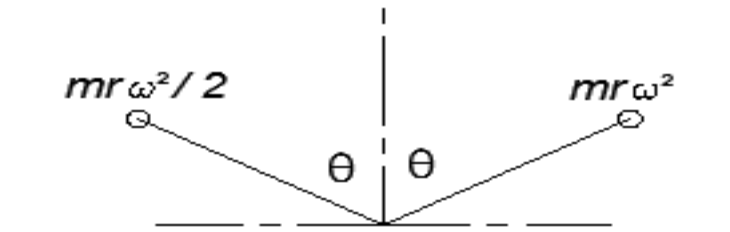
Pr *imary* Direct Force

$$F_{PD} = 2\frac{m}{2}\omega^2 r = m\omega^2 r$$



Pr *imary* Reverse Force

$$F_{PR} = 2\frac{m}{2}\omega^2 r\cos 60 = \frac{m}{2}\omega^2 r$$



*T*otal Horizontal Primary Force

$$F_{PH} = \frac{m}{2}\omega^2 r\sin\theta$$

*T*otal Vertical Primary Force

$$F_{PV} = \frac{3m}{2}\omega^2 r\cos\theta$$

Graphical method (example 1, Solution) cont'd…

Resultant Primary Force

$$F_{PR} = \sqrt{F_{PV}^{\ 2} + F_{PH}^{\ 2}} = 2m\omega^2 r\sqrt{(\frac{3}{4}\cos\theta)^2 + (\frac{1}{4}\sin\theta)^2}$$

as before.

Similarly, the Resultant Secondary Force

$$F_{SR} = \frac{2m\omega^2 r}{n}\sqrt{(\frac{1}{2}\times\frac{\sqrt{3}}{2}\sin 2\theta)^2 + (\frac{1}{2}\times\frac{\sqrt{3}}{2}\cos 2\theta)^2}$$

Example 2

Investigate the out-of-balance forces of an eight-cylinder V-engine consisting of two banks of cylinders, each having four cylinders in line and both working upon one four-throw crankshaft. The centre-lines of the two banks are inclined at angles (∅/2) on each side of the vertical plane. The relative positions of the four cranks are 0°, 180°, 180° and 0° and two connecting rods work on each crank. Find the maximum values of the horizontal and vertical forces acting on the engine, in terms of the angle ∅, the angular velocity ω of the crankshaft, the crank radius r, the connecting rod length l, and the reciprocating mass m per cylinder. State the nature and amount of the total force, (a) when ∅ = 90°, (b) when ∅= 60° (c) ∅ = 45°

Solution (example 2)-analytical method

$$\sum sin\emptyset_i = sin0 + sin180 + sin180 + sin0 = 0$$

$$\sum cos\emptyset_i = cos0 + cos180 + cos180 + cos0 = 0$$

Primary forces are balanced

$$\sum sin2\emptyset_i = sin0 + sin360 + sin360 + sin0 = 0$$

$$\sum cos2\emptyset_i = cos0 + cos360 + cos360 + cos0 = 4$$

Secondary forces are not balanced

$$Total \quad \text{secondary} \quad \text{force} \quad \text{on the} \quad \text{left} \quad \text{bank}$$

$$F_{SL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\theta + 2\alpha) + \sin 2\beta_i \sin(2\theta + 2\alpha)\}$$

$$Total \quad \text{secondary} \quad \text{force} \quad \text{on the} \quad \text{right} \quad \text{bank}$$

$$F_{SR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\alpha - 2\theta) + \sin 2\beta_i \sin(2\alpha - 2\theta)\}$$

Solution (example 2)-analytical method

$$F_{SL} = 4mr\,\omega^2 \cos(2\theta + 2\alpha)$$

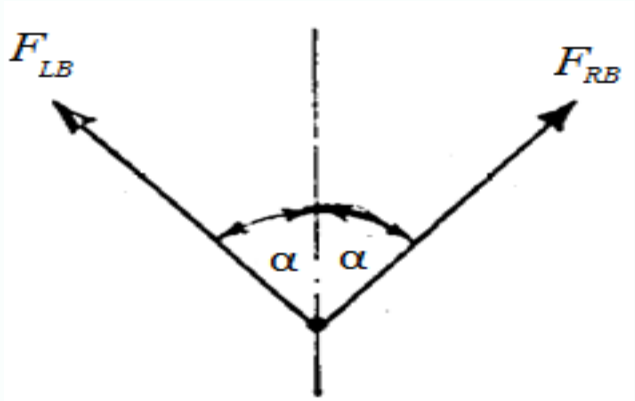
$$F_{SR} = 4mr\,\omega^2 \cos(2\alpha - 2\theta)$$

Total Vertical Force

$$F_V = 8mr\,\omega^2 \cos 2\alpha \cos 2\theta \cos \alpha$$

Total Horizontal Force

$$F_H = 8mr\,\omega^2 \sin 2\alpha \sin 2\theta \sin \alpha$$



Resultant Force

$$F_R = \frac{8mr\,\omega^2}{n}\sqrt{(\cos 2\alpha \cos 2\theta \cos \alpha)^2 + (\sin 2\alpha \sin 2\theta \sin \alpha)^2}$$

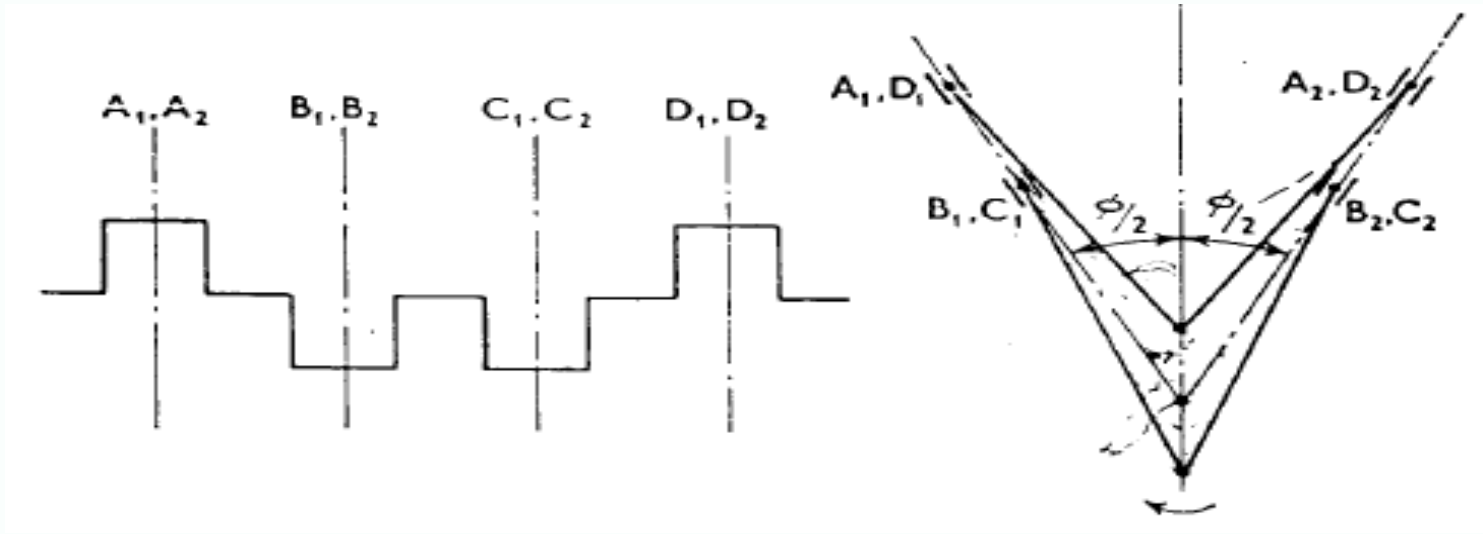
$$\alpha = \frac{\phi}{2}$$

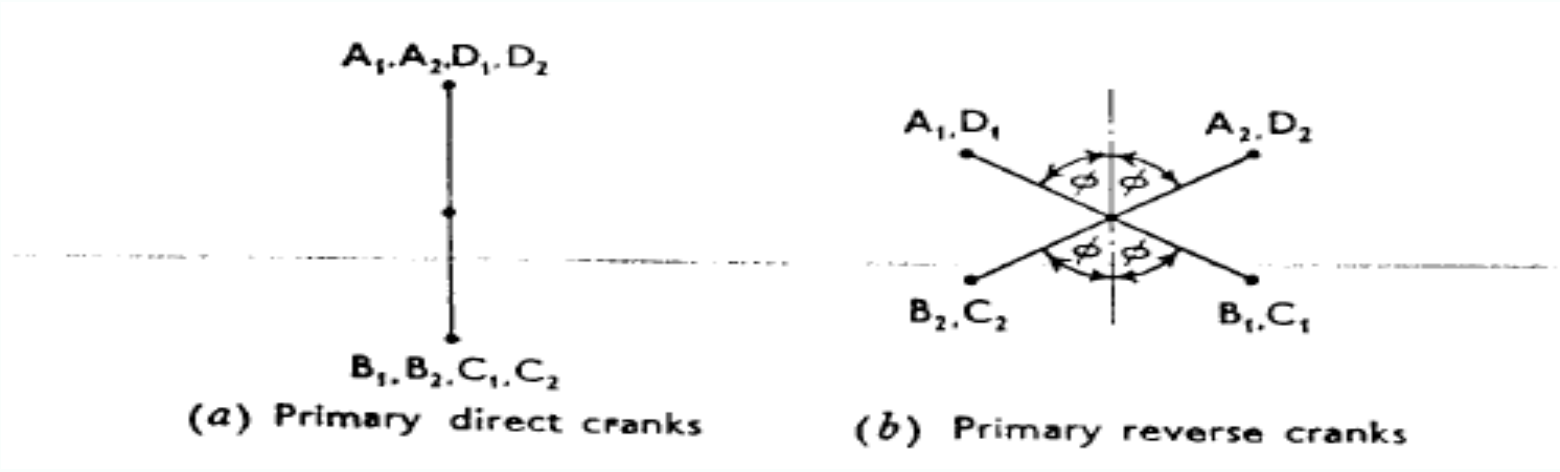
$$when \quad \phi = 90^\circ \qquad F_R = \frac{4\sqrt{2}\, m\omega^2 r^2}{l} \sin 2\theta, \text{ sinusoidal}$$

$$when \quad \phi = 60^\circ \qquad F_R = \frac{2\sqrt{3}\, m\omega^2 r^2}{l}, \text{ constant}$$

$$when \quad \phi = 45^\circ \qquad F_R = ??$$

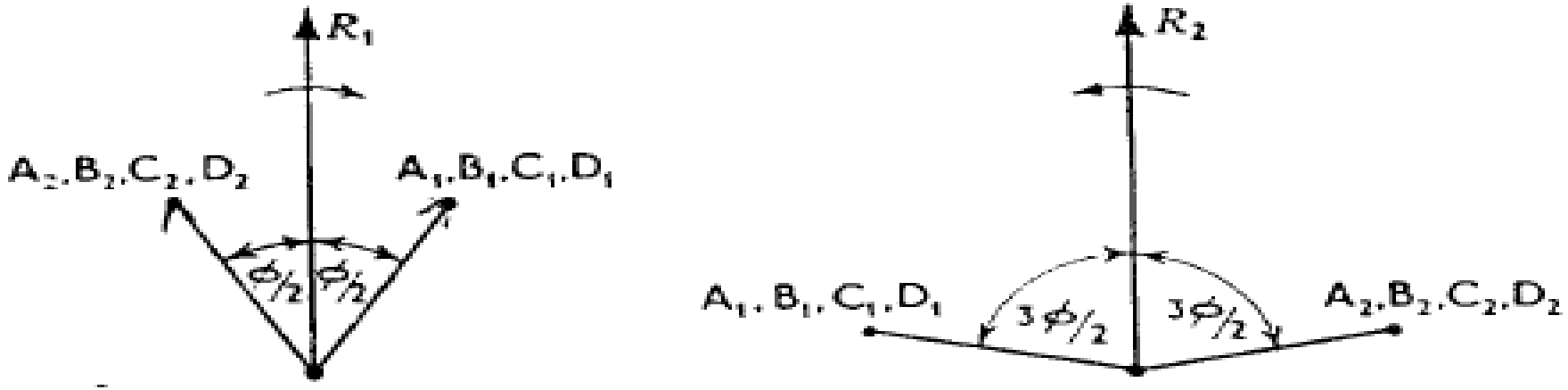
Graphical Method





Graphical Method

Secondary forces. The secondary direct and reverse cranks are shown in Fig. 7.30(*a*) and (*b*).



(*a*) Secondary direct cranks (*b*) Secondary reverse cranks

Fig. 7.30

Graphical Method

Resultant force due to direct cranks, $R_1 = 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{\phi}{2}$

Resultant force due to reverse cranks, $R_2 = 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{3\phi}{2}$

The maximum secondary vertical force occurs when the cranks are vertical, i.e. when R_1 and R_2 are parallel, Fig. 7.31,

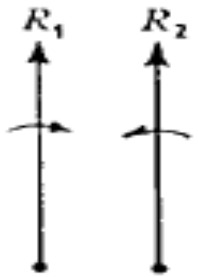


Fig. 7.31



Fig. 7.32

i.e. maximum vertical force

$$= 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{\phi}{2} + 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{3\phi}{2}$$

$$= \frac{8M\omega^2 r^2}{l} \cos\phi \cos\frac{\phi}{2}$$

Graphical Method

The maximum secondary horizontal force occurs when the cranks are at 45° to the vertical, i.e. when R_1 and R_2 are in opposition, Fig. 7.32.
i.e. maximum horizontal force

$$= 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{\phi}{2} - 8 \times \frac{M}{2}\omega^2 \frac{r}{n} \cos\frac{3\phi}{2}$$

$$= \frac{8M\omega^2 r^2}{l} \sin\phi \sin\frac{\phi}{2}$$

When the cranks have rotated through an angle θ from the vertical R_1 and R_2 are inclined at 2θ to the vertical, Fig. 7.33.

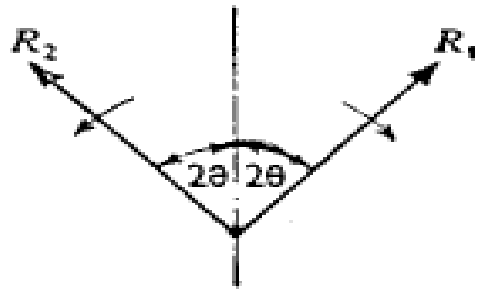


Fig. 7.33

Graphical Method

$$\therefore \text{ vertical force} = (R_1 + R_2)\cos 2\theta$$

$$= \frac{8M\omega^2 r^2}{l}\cos\phi\cos\frac{\phi}{2}\cos 2\theta$$

and $$\text{horizontal force} = (R_1 - R_2)\sin 2\theta$$

$$= \frac{8M\omega^2 r^2}{l}\sin\phi\sin\frac{\phi}{2}\sin 2\theta$$

$\therefore$ resultant force,

$$R = \frac{8M\omega^2 r^2}{l}\sqrt{\left\{\left(\cos\phi\cos\frac{\phi}{2}\cos 2\theta\right)^2 + \left(\sin\phi\sin\frac{\phi}{2}\sin 2\theta\right)^2\right\}}$$

When $\phi = 90°$ $$R = \frac{4\sqrt{2}M\omega^2 r^2}{l}\sin 2\theta$$

When $\phi = 60°$, $$R = \frac{2\sqrt{3}M\omega^2 r^2}{l}$$

In the case when $\phi = 90°$, R varies sinusoidally as the cranks rotate, but when $\phi = 60°$, R is constant.

ASSIGNMENT 1

A shaft carries five masses *A*, *B*, C, D and *E* which revolve at the same radius in planes which are equidistant from one another. The magnitude of the masses in planes *A*, C and D are 50 kg, 40 kg and 80 kg respectively. The angle between *A* and C is 90° and that between C and D is 135°. Determine the magnitude of the masses in planes *B* and *E* and their positions to put the shaft in complete rotating balance.

ASSIGNMENT 2

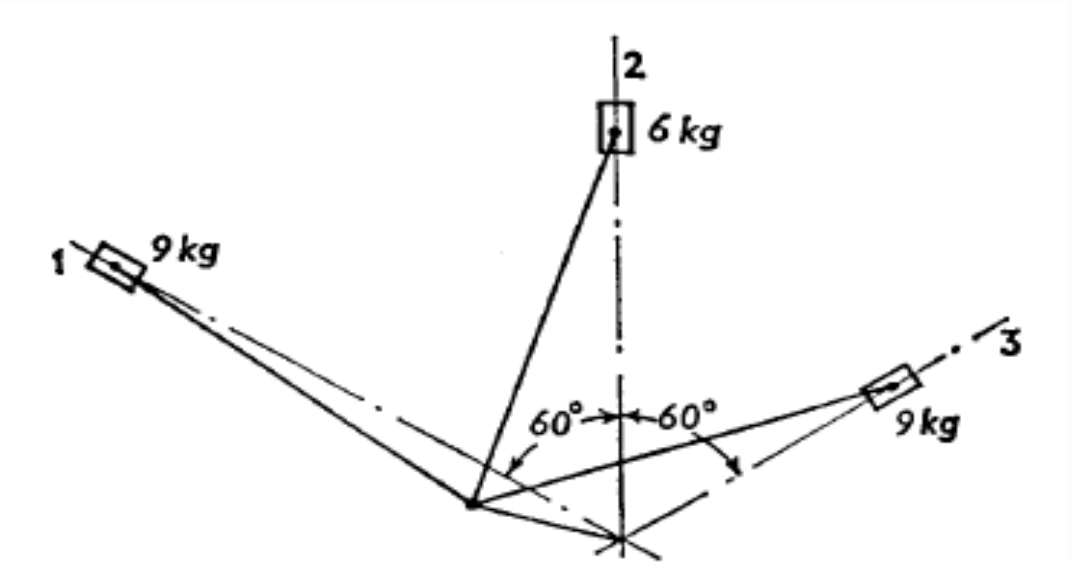
Four masses A, B, C and D are attached to a shaft and revolve in the same plane. The masses are 12 kg, 10 kg, 18 kg and 15 kg respectively and their radii of rotations are 40 mm, 50 mm, 60 mm and 30 mm. The angular position of the masses B, C and D are 60°, 135° and 270° from the mass A. Find the magnitude and position of the balancing mass at a radius of 100 mm.

ASSIGNMENT 3

The spacing of the four cylinders A, B, C and D of a vertical in-line engine is 650 mm, 500 mm and 650 mm. The reciprocating masses of the inner cylinders B and C arc 50 kg, and their cranks are at 60° to one another; the stroke is 325 nun and the connecting rods are 600 mm long. Find the magnitudes of the reciprocating masses for the outer cylinders A and D and the relative angular positions of all the cranks if all primary forces and couples are to be balanced. What will be the maximum unbalanced secondary force acting on the base when the engine is run at 375 rev/min.

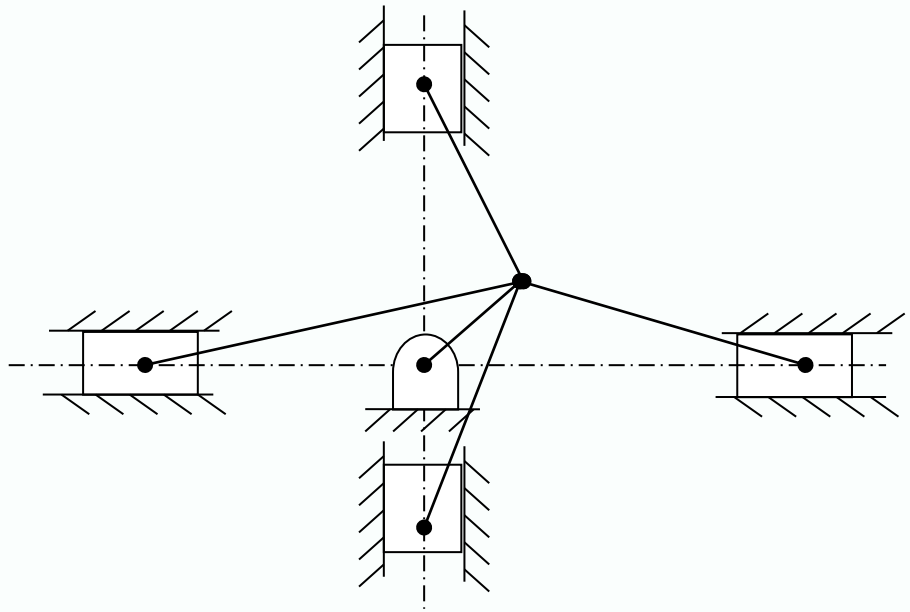
ASSIGNMENT 4

A three-cylinder compressor is arranged as shown with the three pistons operated from a common crank. The reciprocating masses are 9 kg for cylinders 1 and 3, and 6 kg for cylinder 2. The crank radius is 40 mm; and the ratio of connecting rod to crank is 5 for each cylinder. The speed is 960 rev/min. Find the primary unbalanced force at this speed, and obtain a suitable balance mass at crank radius for vertical primary balance. Find also the unbalanced secondary force.



ASSIGNMENT 5

The four-cylinder radial engine depicted in the Figure is an excellent engine from the point of view of dynamic balance. Show that the engine can be balanced by means of a single rotating counterweight mounted on the crankshaft, and determine the magnitude and location of such a counterweight. The crank length is *r*, the connecting rod lengths all equal *l*, and the reciprocating masses all equal *m*. The rotating masses are balanced, and all four cylinders lie in a single transverse plane.



ASSIGNMENT 6

Investigate the state of balance of an eight-cylinder V-engine consisting of two banks of cylinders, each having four cylinders in line and both working upon one four-throw crankshaft. The centre-lines of the two banks are inclined at angles (Ø/2) on each side of the vertical plane. The relative positions of the four cranks are 0°, 90°, 180° and 270° and two connecting rods work on each crank. State the nature and amount of the total force and couple, (a) when Ø = 90°, (b) when Ø= 60° (c) Ø = 45°

THANK YOU